

Inference Under Opposition: Counterfactual Evaluation for Answer Stability in Large Language Models

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Abstract—Understanding whether language models rely on actual meaning or simply exploit annotation artifacts to arrive at the correct conclusion is crucial for developing generalized and robust models. Figuring out the reasoning process of Large Language Models (LLMs) remains a key challenge in explainable AI. In this work, we investigate the robustness of model predictions when faced with counterfactual explanations. We used the Explanations for CommonsenseQA (ECQA) dataset to fine-tune several LLMs. A few-shot generation approach was applied to synthetically produce the counterfactual explanations. To assess the quality of these generated counterfactuals, we introduced the Counterfactual Quality Score (CQS) that measures the merit of the generated counterfactuals. During testing, the fine-tuned model is first presented with a multiple-choice question along with five options and an explanation. After the model produces an initial prediction, a counterfactual explanation is introduced to determine whether its prediction changes. We quantify this sensitivity using an evaluation metric named flip rate, which calculates the percentage where the model’s output flips in response to the counterfactual explanation. Our results highlight the reasoning capabilities of the fine-tuned models and reveal the degree to which model predictions depend on targeted explanations rather than their inherent commonsense understanding.

Index Terms—Natural Language Processing, Large Language Models, Explainable AI, Counterfactual Explanations, Commonsense Reasoning, In-context Learning

I. INTRODUCTION

Behavioral aspects of LLMs have been widely studied [1]. Understanding how models arrive at decisions is key to improving their performance across tasks [2]. Research has therefore evolved from using benchmark datasets for downstream tasks to probing the underlying decision-making processes and explaining model outputs [3].

The Explanations for CommonsenseQA dataset [4], commonly referred to as the ECQA dataset, is designed to study how to elicit responses from LLMs for multiple-choice questions. It focuses on teaching models to extract relevant facts from positive explanations, which contain the correct answer, and to disregard information from negative explanations, which do not contain the correct answer.

The answering of questions in such a dataset, even when the task is extended to open-ended question answering with presented facts, requires the LLMs to focus on the given prompt, context, or explanations to get the right answer [5]. It has been observed that if those explanations are substituted with counterfactual statements, the performance of the LLMs drops even if they are well trained for generalized and specific downstream tasks. The quality of the counterfactuals affects the degradation of the performance [6]. This problem statement is synonymous with designing Chain-of-Thought (CoT) prompting [7], where an LLM is instructed to solve a problem given certain steps and the logical train of information, even though the dataset, like ECQA, has a comparatively less challenging task setup compared to CoT-based question and answering (Q&A).

In our work, we demonstrate that even in the simplest task of answering multiple-choice questions using given options and explanations, LLMs can fail to identify the correct answer when presented with counterfactual explanations. This occurs despite their training to extract facts from positive and negative explanations. We show that some LLMs fine-tuned on the ECQA dataset, with an expectation of developing commonsense, largely fail in test accuracy when their answers are challenged or when they are re-prompted, while others perform much better in the same conditions. That means that different LLM architectures fine-tuned on the same training set exhibit different degrees of stability on test sets. This highlights the need for a thorough evaluation of LLM performance to determine whether they truly learn common sense—i.e., identifying relevant facts and ignoring irrelevant information—or merely rely on spurious associations in the data or dataset artifact to generate answers.

The key contributions of this work are summarized as follows:

- We synthetically generate counterfactual explanations and evaluate the quality of these generated counterfactuals with a novel metric named *Counterfactual Quality Score*

(CQS).

- We demonstrate that even when models are provided with data points along with explanations to learn about common sense reasoning, some of them fail to differentiate between where to extract facts from and where not to.
- We finally validate our observation with the metric, *flip rate*, which calculates the percentage where the fine-tuned model changes its response after being presented with counterfactual explanations.
- We also demonstrate via output-based prompting as well that it is very much possible to derail the LLMs from what it has learned or concluded, raising reasonable doubt about commonsense being injected into these LLMs via dataset orientation.

The investigation to see whether LLMs learn certain behaviors or characteristics, such as *commonsense*, from a dataset is worth pursuing, as it can answer the generalization and commonsense capability of LLMs, especially when it comes to understanding nuanced characteristics of a task.

II. LITERATURE REVIEW

In natural language processing, a counterfactual example refers to a modified version of an input text [8] where minimal edits change the sentiment, classification, or entailment of the output. A counterfactual explanation keeps the context semantically consistent while changing the meaning partially or completely [9]. Such examples are used to test model robustness and reasoning consistency [10].

A. Counterfactual Generation and Analysis of Robustness

Generating counterfactual explanations for training and evaluating LLMs is challenging. [11] highlight that while human annotations are valuable, explanation quality can be subjective, and [8] note the high cost of manual annotation. To address this, we synthetically generate counterfactual explanations and assess their quality using a novel metric.

As LLMs see widespread deployment, robustness becomes critical; counterfactual testing offers an effective way to evaluate it [12], making high-quality synthetic counterfactual generation essential [8].

[13] propose two LLM-based approaches: (i) inference-time targeted edits that preserve confounding concepts, and (ii) guided training-time instructions to better structure the embedding space, with the latter being more resource-efficient and effective. Their model-agnostic design enables broad applicability.

[14] introduce a human-in-the-loop, interactive counterfactual framework where users iteratively refine perturbations to probe model behavior. This approach uncovers diverse failure modes (e.g., surface-level vs. semantic errors) and supports deeper analysis of model sensitivities, advocating its integration into robustness and interpretability workflows.

B. Explanation-Enhanced Models for CommonsenseQA

[4] introduce the ECQA dataset, which augments CommonsenseQA with structured, human-annotated explanations split into positive facts (supporting correct answers) and negative facts (refuting distractors). Covering 11,000 QA pairs, it

enables tasks beyond answer prediction, such as explanation generation and evaluation of underlying reasoning process.

[15] build on this idea with a discussion-based framework (CMD), showing that a single well-prompted agent can often match multi-agent systems.

[16] examine whether strong benchmark performance reflects robust reasoning or memorization. Using ECQA, they show that models like ChatGPT often fail to identify and apply the relevant commonsense knowledge, suggesting current benchmarks may overestimate reasoning ability and confuse knowledge access with reasoning skill.

III. METHODOLOGY

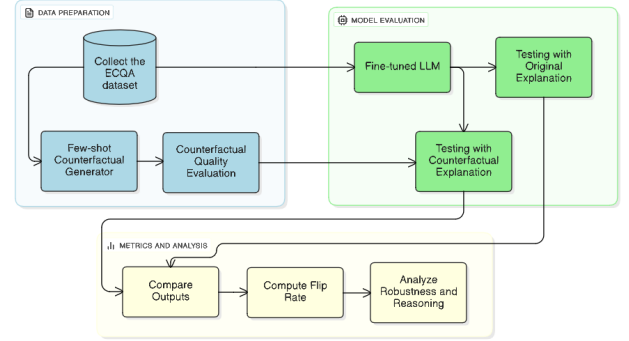


Fig. 1: End-to-End Pipeline for Robustness Testing with Counterfactual Explanations.

A. Overview

The pipeline uses a subset of the ECQA dataset with human annotations to define counterfactuals based on properties outlined in subsection III-D. These annotations guide in-context generation of counterfactual examples, which are then evaluated using defined metrics. We use these examples to measure performance drops in fine-tuned LLMs and assess their stability via *flip rate*. Further analysis probes whether LLMs truly learn commonsense reasoning by exposing them to misleading or counterfactual explanations and prompt variations, motivating reasonable doubt about their robustness. Overall, we examine whether models develop genuine commonsense or rely on pattern extraction during fine-tuning.

Commonsense in LLMs, for this task, is the ability to extract correct facts from valid explanations and reject incorrect ones, thereby eliminating wrong answer choices. Failures in doing so can specifically point to robustness issues—such as architectural or stability limitations—rather than an absence of commonsense reasoning.

B. Dataset

The ECQA dataset introduced by [4] forms the basis of this work. Each entry contains a question in `q_text`, five options (`q_op1–q_op5`), and the correct answer in `q_ans`. It also includes explanations: `taskA_pos` justifies the correct answer, while `taskA_neg` explains why the other options are incorrect. A snapshot of the ECQA dataset is illustrated in the relevant rows of Table I.

Field	Content
Question (q_text)	I framed the picture so I could look at it while I ate dinner, so where did I put it?
Options (q_op1 to q_op5)	newspaper, desktop, art show, art museum, table
Correct Answer (q_ans)	table
Positive Explanation (taskA_pos)	Frames are usually kept on a flat surface so they can be seen. People eat dinner on the table. If one wants to look at the picture while eating, they tend to put it on the table.
Negative Explanation (taskA_neg)	Newspapers are usually not kept on the dinner table. One cannot look at the picture while eating if it is on the desktop. People do not eat dinner at an art show. People do not eat dinner at an art museum.
Positive Counterfactual (taskA_pos_CF)	Newspaper is used to read news. People read newspapers when they are eating. If one wants to look at the picture while eating, they tend to put it on the newspaper.
Negative Counterfactual (taskA_neg_CF)	Framed pictures are usually not kept on a table. One cannot look at the picture while eating if it is on the desktop. People do not eat dinner at an art show. People do not eat dinner at an art museum.

TABLE I: Snapshot of the dataset along with generated counterfactual explanations.

Model	Total Parameters	Trainable Parameters
Mistral-7B-Instruct-v0.3	3,758,362,624	268,701,696
Meta-Llama-3-8B-Instruct	4,540,600,320	1,050,939,392

TABLE II: Details of the fine-tuned models.

C. Counterfactual Generation

Utilizing the question, answer options, correct answer, and positive & negative explanations from the dataset, we generated the counterfactual explanations using our proposed approach outlined in algorithm 1. The taskA_pos_CF and taskA_neg_CF rows in Table I include examples of our generated counterfactual explanations.

Algorithm 1 Counterfactual Generation via In-Context Learning

Require: Dataset $D = \{(Q_i, O_{ij}, A_i, PE_i, NE_i)\}, i = 1, \dots, N, j = 1, \dots, 5$
Require: Few-shot examples $FE = \{(Q, O, A, PE, NE, PE^{CF}, NE^{CF})\}_{k=1}^K$
Ensure: Counterfactual dataset D_{CF}
1: $D_{CF} \leftarrow \emptyset$
2: **for all** $(Q, O_j, A, PE, NE) \in D$ **do**
3: Sample K examples from FE to form context
4: $(PE^{CF}, NE^{CF}) \leftarrow \text{CF_Generate}(Q, O_j, A, PE, NE)$
5: $(CQS_P, CQS_N) \leftarrow \text{ComputeCQS}(PE^{CF}, NE^{CF})$ ▷ See Eq. 4
6: $\text{entry} \leftarrow (Q, O_j, A, PE, PE^{CF}, NE, NE^{CF}, CQS_P, CQS_N)$
7: Append entry to D_{CF}
8: **end for**
9: **return** D_{CF}

Here, Q refers to the questions, O_j are the five options, A is the correct answer, PE and NE are the positive and negative explanations, respectively. The generated positive and negative counterfactual explanations are designated by PE_{CF} and NE_{CF} in the algorithm. By computing the CQS , we were able to determine the quality of the generated output. Counterfactual explanations with a CQS score above 0.8 are considered high quality. Those with scores between 0.7 and 0.8 are categorized as average quality, while explanations with a CQS below 0.7 are classified as low quality.

D. Quality Evaluation of Counterfactuals

To determine the quality of the counterfactual explanations we were producing, first, we had to define the properties of a good counterfactual explanation. In the context of our research, a good counterfactual explanation should:

- Be semantically close to the original (retains structure, tone, or logic pattern) explanation.

- Be lexically plausible given the question context.
- Flip the reasoning outcome, such that the correct answer no longer makes sense, but a wrong one now does.

We generated counterfactual explanations for the positive as well as negative explanations found in the dataset (taskA_pos and taskA_neg). Both types of counterfactual explanations had to be semantically similar to the original explanations. For each explanation E , generated counterfactual C , Correct Answer Ans , and Options Ops , We calculated:

1) Similarity score:

$$S_{sim}(E, C) = \cos(\text{emb}(E), \text{emb}(C)) \quad (1)$$

which measures how semantically close C is to E . The cosine similarity index has a range of $[-1, 1]$, which was rescaled to $[0, 1]$ for our purpose. The rescaled score is:

$$\tilde{S}_{sim} = \frac{S_{sim} + 1}{2} \quad (2)$$

- #### 2) Flip / Correctness score:
- The correct answer (q_ans) should not appear in the generated positive counterfactual explanation, and at least one of the wrong options ($q_op1 - q_op5$, excluding q_ans) should appear. On the other hand, the correct answer must be present in the negative counterfactual explanation, replacing one of the wrong options. This ensures the counterfactual explanations actually flip the reasoning, i.e., change which answer makes sense.

$$S_{flip}(C, Ans, Ops) = \alpha \cdot A + (1 - \alpha) \cdot B \quad (3)$$

The values of A and B differ for positive and negative counterfactuals. For positive counterfactuals, ($A = 1$) if C does not include the correct answer, and ($B = 1$) if (C) includes a wrong option (1–5). For negative counterfactuals, ($A = 0$) if (C) excludes the correct answer, and ($B = 1$) if the correct answer replaces a wrong option. In both cases, ($\alpha = 0.6$).

- #### 3) We combine (2) and (3) to a single Counterfactual Quality Score (CQS).

$$CQS = w_{sim} \cdot \tilde{S}_{sim} + w_{flip} \cdot S_{flip} \quad (4)$$

Where, $w_{sim} = w_{flip} = 0.5$. CQS ranges from $[0, 1]$, and a higher score indicates a better counterfactual.

E. Model Fine-tuning and Setup

Two pretrained models (*Meta-Llama-3-8B-Instruct* and *Mistral-7B-Instruct-v0.3*) from the Huggingface API Library or huggingface_hub were chosen to be fine-tuned. The models were fine-tuned in 4-bit quantization with Low-Rank Adaptation (LoRA) due to the resource-heavy nature of the LLMs. The number of total and trainable parameters of these models is described in Table II.

F. Framework

Figure 1 outlines our pipeline. We begin with the ECQA dataset, using human annotations to create few-shot counterfactual examples. These are used to synthetically generate counterfactual explanations for the test set, while the training and validation sets are used to fine-tune the language models. Evaluation follows a two-phase process: first, models generate initial responses to questions and answer options, yielding a baseline accuracy. In the second phase, models are re-prompted with counterfactual explanations for the same questions to assess consistency or response flipping, after which post re-prompting accuracy is computed.

IV. EVALUATION METRICS

Accuracy: the proportion of predictions the model gets correct.

F1 Score: the harmonic mean of precision and recall, balancing false positives and false negatives.

Flip Rate: the percentage of predictions that change after introducing a counterfactual explanation, indicating model sensitivity.

Algorithm 2 Flip-Rate Evaluation of Fine-tuned LLM

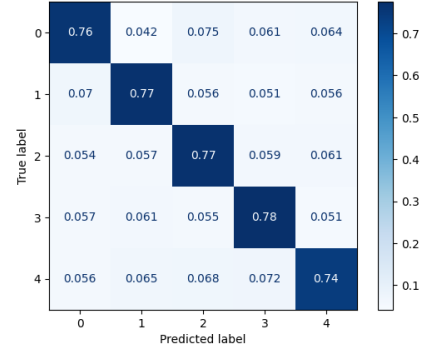
Require: Fine-tuned Model M , Counterfactual Dataset D_{CF}
Ensure: Flip Rate F

- 1: Initialize $flip_count \leftarrow 0$
- 2: **for** each example (Q, O, E, E_{CF}, A) in D_{CF} **do**
- 3: Predict with original explanation:
 $A_{orig} \leftarrow M(Q, O, E)$
- 4: Re-prompt with counterfactual explanation: $A_{CF} \leftarrow M(Q, O, E_{CF})$
- 5: **if** $A_{orig} \neq A_{CF}$ **then**
- 6: $flip_count \leftarrow flip_count + 1$
- 7: **end if**
- 8: **end for**
- 9: Compute $F \leftarrow flip_count / |D_{CF}|$
- 10: **return** F

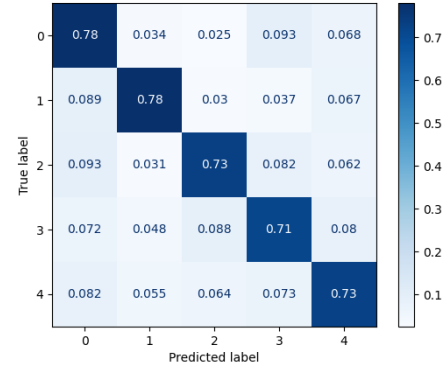
V. RESULT ANALYSIS

Models	Mistral-7B		Llama-8B	
	Accuracy (%)	F1-Score (%)	Accuracy (%)	F1-Score (%)
Baseline	76.2	76.2	73.9	73.8
Prompted with positive counterfactuals	76.2	76.1	71.0	70.8
Prompted with positive and negative counterfactuals	76.2	76.1	71.12	71.1

TABLE III: Performance comparison of models under different prompting strategies.



(a) Baseline of Mistral-7B.



(b) Baseline of Llama-8B.

Fig. 2: Normalized Confusion Matrices of Both Fine-tuned Models.

The fine-tuned models were evaluated on the ECQA test set (Table III). Using correct explanations as a baseline, Mistral-7B achieved 76.2% accuracy and 76.1% F1-score, maintaining consistent performance across all three settings: correct explanations, synthetically generated positive explanations, and combined positive-negative counterfactuals.

In contrast, Llama-8B showed less stability, with accuracy dropping from 73.9% (correct explanations) to 71.0% and 71.12% when tested with positive-only and combined counterfactual explanations, respectively.

We further evaluated whether counterfactual explanations could induce prediction flips. In this setup, models first selected one of five answer options (0–4). We then re-prompted them with a counterfactual explanation and asked if they would change their answer. Counterfactuals were generated using *GPT-3.5 Turbo* and *Meta-Llama-3-8B-Instruct*.

GPT-generated counterfactuals achieved higher quality, with average CQS scores of 0.86 (positive) and 0.92 (negative), compared to Llama-generated explanations with scores of 0.61 (positive) and 0.65 (negative), respectively.

Mistral-7B showed stable behavior across both high-quality (GPT-generated, *Average CQS* > 0.8) and low-quality (Llama-generated, *Average CQS* < 0.7) counterfactuals, with an overall flip rate of 11.08% (243/2194). Despite this sensitiv-

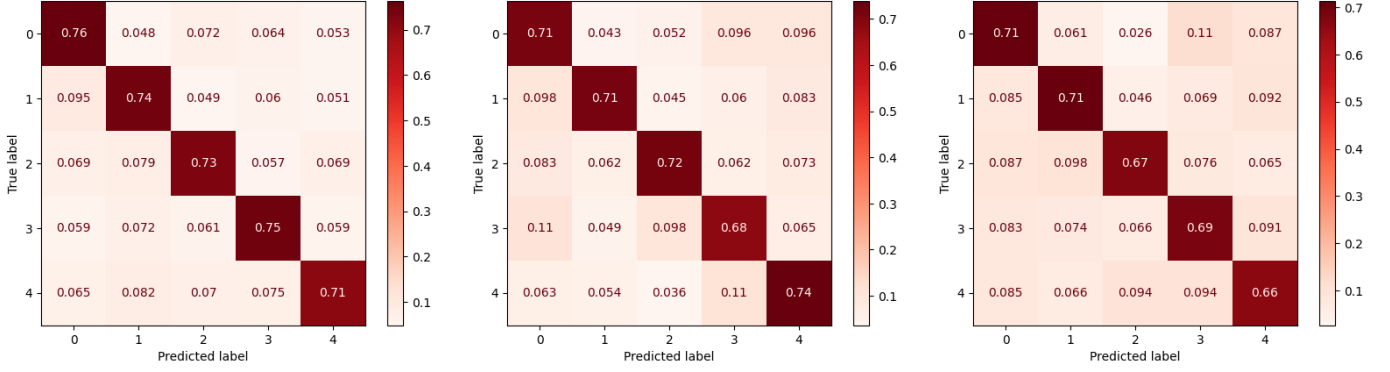


Fig. 3: Normalized Confusion Matrices of Both Models after Re-prompting.

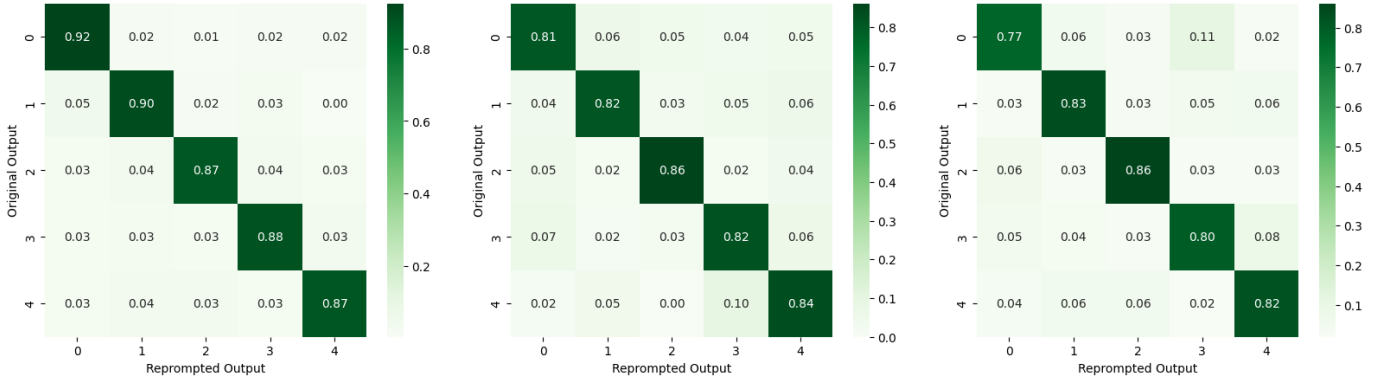


Fig. 4: Normalized Flip Matrices of Both Fine-tuned Models.

ity, performance degradation was minimal: accuracy dropped only from 76.2% to 73.9% and F1-score from 76.1% to 73.8%.

A detailed breakdown shows that 5.74% of cases flipped from correct to incorrect, 3.42% from incorrect to correct, and 1.91% from incorrect to incorrect. This explains why the overall flip rate does not translate into a substantial accuracy drop.

In contrast, Llama-8B showed less consistent behavior. With GPT-generated counterfactuals, it achieved a flip rate of 21.67%, increasing to 26.0% with Llama-generated explanations. For high-CQS counterfactuals, the model flipped 474 times, including 10.3% correct-to-incorrect changes (226 cases) and 6.24% incorrect-to-correct changes (137 cases), resulting in a 4.06% accuracy drop (73.8% $\rightarrow$ 69.74%).

For low-CQS counterfactuals, 571 total flips were observed, with 282 correct-to-incorrect and 154 incorrect-to-correct transitions. Results for both models are summarized in Table IV.

Figures 2, 3, and 4 highlight the performance of both models across the five answer classes (0–4). Figure 2 illustrates the baseline normalized confusion matrices, while figure 3

includes the matrices after re-prompting. As the Mistral-7B model performed consistently for both high and low quality counterfactual explanations, it has the same confusion matrix after re-prompting as well as the same flip matrix, which are displayed in figures 3a and 4a respectively. In contrast, the Llama-8B model had different outcomes depending on the quality of counterfactual explanations. That is why, there are distinct confusion as well as flip matrices for Llama-8B shown through figures 3b, 3c and 4b, 4c.

Models	Mistral-7B		Llama-8B	
Quality of Counterfactual Explanations	High	Low	High	Low
Total Flipped	243	243	474	571
Flip Rate	11.08%	11.08%	21.6%	26.02%
Right $\rightarrow$ Wrong	126	126	226	282
Wrong $\rightarrow$ Right	75	75	137	154
Other (Both wrong)	42	42	111	135

TABLE IV: Flip statistics of Mistral-7B and Llama-8B.

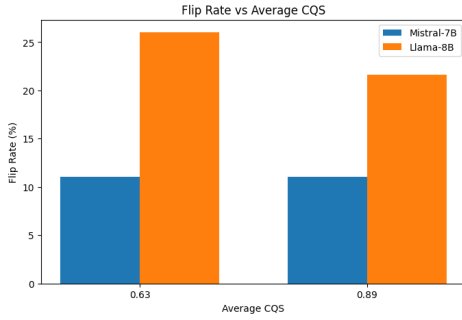


Fig. 5: Flip Rate vs Average CQS.

VI. DISCUSSION

Overall, the analysis reveals notable differences in how the two models respond to counterfactual explanations. Mistral-7B maintains a stable baseline accuracy of 76.2% across positive, and combined positive–negative counterfactual settings, showing consistent behavior regardless of counterfactual quality (high or low CQS), suggesting limited sensitivity to explanation variations. In contrast, despite having more parameters, Llama-8B performs worse across all evaluation stages, indicating weaker robustness to counterfactual perturbations.

Figure 5 compares model stability across counterfactual types. Mistral-7B maintains a consistent flip rate of 11.08% for both high- and low-quality counterfactuals, suggesting relatively robust behavior and reduced sensitivity to explanation quality.

In contrast, Llama-8B shows higher and more variable flip rates (21.6% for high-quality and 26.02% for low-quality counterfactuals), indicating stronger sensitivity to counterfactual inputs. Its lower accuracy and fluctuating F1-score further suggest reliance on superficial patterns or dataset artifacts rather than robust reasoning.

VII. CONCLUSION & FUTURE WORKS

This work evaluates LLM robustness and reasoning consistency by examining how predictions change under counterfactual explanations. Using the ECQA dataset, we generate high- and low-quality counterfactuals via few-shot (in-context) learning and measure sensitivity through *flip rate*. Results show clear differences: Mistral-7B exhibits relatively stable behavior with low flip rates, while Llama-8B is more sensitive, especially to low-quality counterfactuals—likely due to their weaker linguistic signals, which hinder distinguishing valid from invalid explanations. The variability in Llama-8B’s performance suggests reliance on superficial cues rather than commonsense reasoning. However, Mistral-7B’s stability should be interpreted cautiously, as it may reflect retained dataset patterns despite counterfactual perturbations, even though test instances were unseen during training.

Overall, this study presents a systematic framework for generating and evaluating counterfactual explanations, introduces *flip rate* as a robustness metric, and demonstrates its usefulness in analyzing LLM reasoning stability. Our results highlight the

need to assess not only accuracy but also consistency under counterfactual perturbations.

Future work should investigate architectural factors underlying Mistral-7B’s relative stability compared to Llama-8B, despite the latter’s larger size, and explore broader counterfactual testing setups. We also release a counterfactual version of the ECQA dataset in an anonymized repository here.

VIII. ETHICAL CONSIDERATION

The ethical considerations primarily concern fine-tuning LLMs for counterfactual generation or using in-context examples, where model biases or differing interpretations could complicate comparisons. However, in this work, counterfactuals are explicitly defined and generated from aligned annotated examples, mitigating such concerns. Therefore, the study does not introduce inherent ethical issues in model evaluation.

IX. ACKNOWLEDGMENT

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